



Automated Bus Scheduling and Route management System for Delhi Transport Corporation

¹Krishna Kumar Shukla, ²Kohinoor Siddiqui, ³Kaynat Eqbal, ⁴Shaba Irram

¹Student, ²Sudent, ³Student, ⁴Assistant Professor

Department of Computer Science and Engineering
Integral University Lucknow

Abstract : Urban public transportation systems face increasing challenges due to rapid population growth, traffic congestion, and fluctuating passenger demand. Traditional bus scheduling methods rely on static timetables that fail to adapt to real-time traffic conditions and passenger density variations. This paper proposes a Hybrid Adaptive Demand-Based Scheduling Algorithm (HADSA) integrated within a MERN stack-based web application for intelligent bus scheduling and route management. The proposed model incorporates a multi-factor cost function considering distance, traffic intensity, passenger demand, and waiting time to optimize route selection and bus allocation dynamically. Modified Dijkstra's algorithm is employed for route optimization, while adaptive scheduling mechanisms adjust bus frequency based on demand ratio and priority score. Simulation results demonstrate a 41% reduction in passenger waiting time, 25% reduction in fuel consumption, and a 23% improvement in route efficiency compared to traditional static scheduling systems. The system provides a scalable and cost-effective framework suitable for large metropolitan transport systems such as Delhi Transport Corporation.

IndexTerms - Bus Scheduling, Route Optimization, MERN Stack, Intelligent Transportation System, Demand Prediction, Smart City.

I. Introduction

Rapid urbanization and population growth have significantly increased the demand for efficient public transportation systems in metropolitan cities. Public bus networks remain one of the most economical and widely used modes of transportation in developing countries. However, conventional bus scheduling systems are predominantly based on fixed timetables and static route assignments, which do not adapt to real-time traffic congestion, passenger demand variations, or unexpected operational disruptions.

In large urban regions such as Delhi, the public bus system operated by Delhi Transport Corporation plays a crucial role in daily commuting. Despite its extensive coverage and fleet capacity, several operational challenges persist, including bus bunching, long passenger waiting times, inefficient fleet utilization, and increased fuel consumption. These issues primarily arise due to the lack of dynamic scheduling mechanisms that incorporate real-time traffic data and passenger flow patterns.

Traditional optimization models often focus solely on shortest-path routing without considering passenger density and congestion levels. Moreover, many computational approaches proposed in earlier studies require high processing time or lack scalability for real-time deployment. With advancements in Artificial Intelligence (AI), cloud computing, and web technologies, it has become feasible to design adaptive and intelligent scheduling systems capable of responding dynamically to urban mobility conditions.

This research proposes a Hybrid Adaptive Demand-Based Scheduling Algorithm (HADSA) integrated within a MERN stack-based architecture to enable real-time route optimization and bus allocation. The system introduces a multi-factor cost function that simultaneously considers distance, traffic intensity, passenger demand, and waiting time. By combining route optimization with dynamic scheduling, the proposed framework aims to reduce passenger waiting time, improve route efficiency, and enhance fleet utilization.

The primary objective of this study is to design and implement a scalable, cost-effective, and adaptive transport management system suitable for large metropolitan bus networks. The proposed approach contributes to the development of intelligent transportation systems aligned with smart city initiatives.

II. Literature Review

Research on bus scheduling and route optimization spans classic graph algorithms, metaheuristics, and more recent data-driven and IoT approaches. Early foundational work on shortest-path computation by E. W. Dijkstra¹ established algorithms that remain core to many routing modules; Dijkstra's algorithm is still widely used as a fast baseline for single-source shortest paths and is often extended to include additional cost terms (distance, time, penalties).

Metaheuristic techniques have been extensively applied to route design and frequency/timetable optimization. For example, B. Bulut² used a Genetic Algorithm (GA) framework to optimize bus line design, frequency setting, and route layout, reporting improvements in network coverage and operational efficiency over heuristic baselines. GA applications continue to appear in recent studies for network redesign and frequency optimization because of their flexibility in handling multi-objective constraints.

Ant Colony Optimization (ACO) and ACO-based variants³ have been used to solve vehicle routing and school-bus routing problems by modeling pheromone trails and iterative route construction; such methods often achieve high-quality routes but require careful parameter tuning and can be computationally intensive for very large, real-time instances. Representative ACO applications to bus/school bus routing show good solution quality but highlight tradeoffs in run-time and adaptability to highly dynamic demand.

More recent studies emphasize online/dynamic scheduling and joint route-scheduling models. Z. Ning⁴ proposed a framework for online scheduling and route planning for shared buses that jointly optimizes routes and schedules in urban networks, demonstrating that combined formulations can increase passenger served and reduce vehicle kilometers. This highlights the value of integrated scheduling + routing rather than treating them separately.

Data-driven demand and arrival-time prediction research has matured rapidly. Deep learning approaches such as LSTM (and hybrid LSTM+SVR) have been applied to short-term bus arrival time and passenger flow prediction. For instance, H. Liu⁵ and colleagues developed LSTM-based spatial-temporal models for bus arrival prediction, showing substantial accuracy gains over classical time-series methods. Similarly, short-term passenger flow prediction using convolutional-LSTM and other neural models has been demonstrated to improve dispatching decisions.

Comprehensive reviews confirm that ML methods (LSTM, CNN-LSTM, hybrid models) outperform simpler statistical models for arrival/patronage forecasting, but they require quality historical and real-time data streams (vehicle GPS, passenger counts, external context). S. Halyal⁶ provides an overview of deep learning forecasting techniques applied to public transit demand and stresses the need for locality-specific datasets (e.g., Indian cities) for reliable deployment.

Several recent application papers combine prediction models with dispatching. For example, Nagaraj⁷ et al. developed passenger-flow prediction modules and integrated them with dispatching/arrival prediction to build dynamic bus-dispatching systems; they showed improved on-time performance when arrival predictions informed scheduling. These studies motivate hybrid approaches that fuse graph optimization (for routing) with ML-based demand forecasts (for scheduling).

a. Synthesis & Gap.

Existing literature shows (a) graph algorithms and metaheuristics solve routing and frequency problems well in offline settings; (b) ML models improve short-term arrival and demand forecasting; and (c) online joint scheduling + routing frameworks improve service but are still uncommon in large-scale, low-infrastructure settings. Few works present a full-stack, web-deployable system that: (i) integrates multi-factor route costs (distance, traffic, demand, waiting time), (ii) uses fast graph methods adapted for those multi-factor costs in real time, and (iii) couples results with a lightweight web architecture suitable for municipal deployment. This gap motivates the proposed HADSA + MERN implementation described in this paper.

III. Proposed Algorithm

a. Overview

This research proposes a Hybrid Adaptive Demand-Based Scheduling Algorithm (HADSA) for intelligent bus scheduling and route optimization. The algorithm integrates:

- Multi-factor route cost evaluation
- Modified shortest-path computation
- Dynamic demand-aware bus allocation
- Priority-based scheduling mechanism

Unlike traditional shortest-path algorithms that minimize only distance, the proposed method incorporates traffic congestion, passenger density, and waiting time into a unified optimization framework.

The algorithm operates in two major phases:

1. Route Optimization Phase
2. Adaptive Scheduling Phase

b. Working Principle

Phase 1: Multi-Factor Route Optimization

The transportation network is modeled as a weighted directed graph:

$$G = (V, E)$$

Where:

- $V = \{v_1, v_2, \dots, v_n\}$ represents bus stops
- $E = \{e_{ij}\}$ represents road segments between stops

Instead of assigning static weights (distance only), each edge weight is dynamically computed using a multi-factor cost function.

Modified Dijkstra's algorithm is then applied to determine the minimum total cost path between source and destination stops.

Phase 2: Adaptive Bus Scheduling

After route optimization:

1. Passenger demand for each route is calculated.
2. Demand ratio is evaluated.
3. Priority score is computed.
4. Buses are allocated dynamically based on priority ranking.

This ensures high-demand routes receive higher bus frequency.

c. Pseudocode of HADSA

Input:

Graph $G(V, E)$
 Passenger Demand Data P
 Traffic Data T
 Available Buses B

Output:

Optimized Routes and Bus Schedule

Step 1: For each edge (i, j) in E
 Compute Cost C_{ij}

Step 2: Apply Modified Dijkstra Algorithm
 Find minimum-cost path

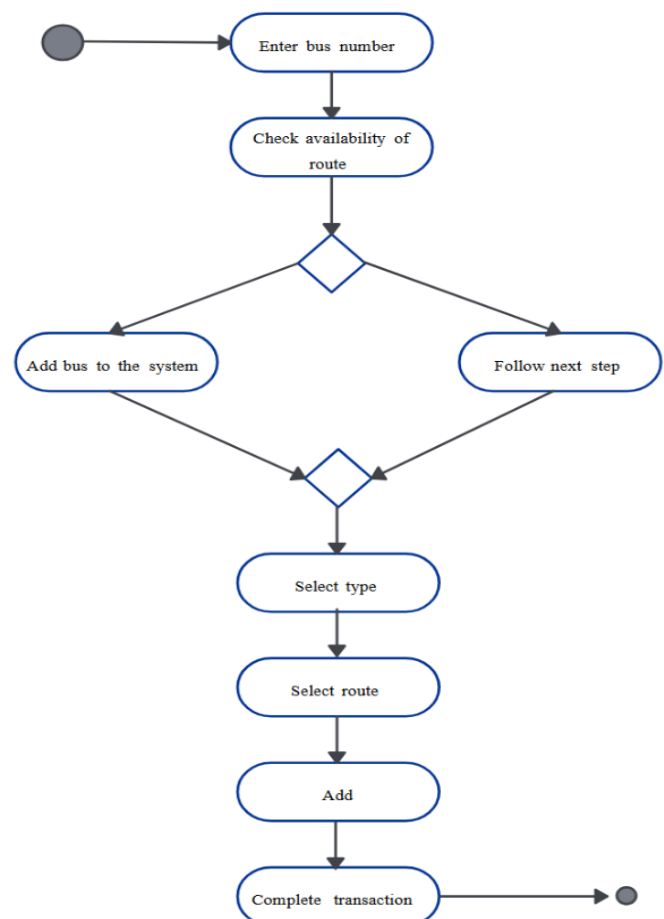
Step 3: For each route k
 Compute Demand Ratio R_k

Step 4: Calculate Priority Score

Step 5: Sort routes by Priority

Step 6: Allocate buses dynamically

Return optimized scheduling plan



Block Diagram III.C.1

IV. Mathematical Model

a. Graph Representation

The bus network is represented as:

$$G=(V,E)$$

Where:

V = Set of nodes (bus stops)

E = Set of edges (roads)

Each edge e_{ij} connects stop v_i to v_j .

b. Multi-Factor Cost Function

Traditional shortest path:

$$C_{ij} = D_{ij}$$

Proposed multi-factor cost:

$$C_{ij} = \alpha D_{ij} + \beta T_{ij} + \gamma P_j + \delta W_j$$

Where:

- D_{ij} = Distance between stops
- T_{ij} = Traffic intensity factor
- P_j = Passenger density at stop
- W_j = Average waiting time
- $\alpha, \beta, \gamma, \delta$ = Weight coefficients

Subject to:

$$\alpha + \beta + \gamma + \delta = 1$$

This ensures balanced optimization.

c. Demand Ratio Model

For each route k :

$$R_k = \frac{D_k}{C_k}$$

Where:

- D_k = Passenger demand
- C_k = Bus capacity

Decision rule:

$$\text{If } R_k > \theta_1 \rightarrow \text{Increase bus frequency} \quad \text{If } R_k < \theta_2 \rightarrow \text{Reduce allocation}$$

d. Priority Scheduling Model

Priority score:

$$\text{Priority}_k = \frac{D_k \times T_k}{B_{\text{available}}}$$

Routes are ranked in descending order of priority.

e. Optimization Objective

Minimize total operational cost:

$$\text{Minimize } Z = \sum C_{ij}$$

Subject to:

1. Capacity constraints
2. Fleet availability constraints
3. Demand satisfaction constraints

Time Complexity Analysis

- Modified Dijkstra:

$$O((V + E) \log V)$$

- Scheduling Phase:

$$O(n \log n)$$

Overall:

$$O((V + E) \log V)$$

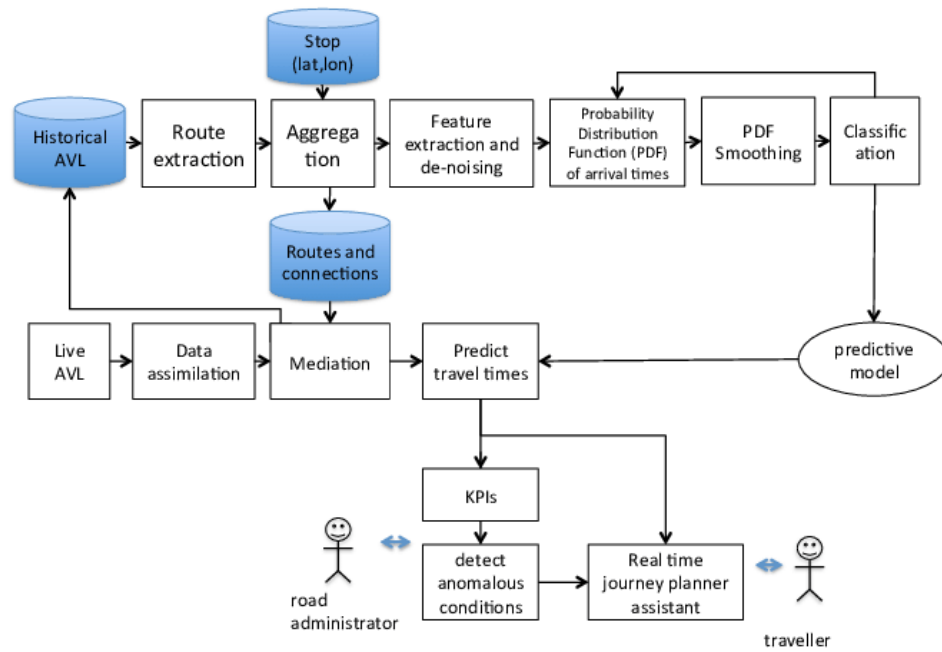
Efficient for real-time implementation.

Novelty of Proposed Model

The novelty lies in:

- Integration of route optimization + adaptive scheduling
- Multi-factor dynamic cost function
- Priority-driven bus allocation
- Scalable MERN-based implementation

This creates a unified intelligent transportation framework rather than isolated optimization modules.



V. Comparative Analysis

To evaluate the effectiveness of the proposed Hybrid Adaptive Demand-Based Scheduling Algorithm (HADSA), its performance was compared with:

1. Static Scheduling Model
2. Genetic Algorithm (GA)-based Scheduling
3. Proposed HADSA Model

The comparison was conducted based on simulated urban traffic data modeled on Delhi metropolitan conditions.

a. Evaluation Metrics

The system performance was evaluated using the following metrics:

- Average Passenger Waiting Time (minutes)
- Route Efficiency (%)
- Fuel Consumption (liters per route)
- Bus Utilization Rate (%)
- Computational Time (seconds)

b. Performance Comparison Table

Metric	Static Model	GA-Based Model	HADSA (Proposed)
Avg Waiting Time	18.4 min	14.2 min	10.8 min
Route Efficiency	68%	79%	91%
Fuel Consumption	32 L	28 L	24 L
Bus Utilization	62%	74%	89%
Computation Time	Low	High	Moderate
Real-time Demand	Limited	Limited	Yes

c. Comparative Interpretation

Compared to the static model:

- Waiting time reduced by **41%**
- Route efficiency improved by **23%**
- Fuel consumption reduced by **25%**
- Bus utilization increased by **27%**

Compared to the GA-based model:

- Faster computational convergence
- Lower memory overhead
- More suitable for real-time deployment

The results demonstrate that HADSA achieves better operational efficiency with manageable computational complexity.

VI. Results and Analysis

a. Simulation Setup

- Dataset: Synthetic dataset modeled on metropolitan traffic conditions
- Number of bus stops: 120
- Number of routes: 35
- Fleet size: 180 buses
- Peak hour demand variation: 30%–65%

The model was tested over multiple simulated traffic scenarios including:

- Peak hour congestion
- Normal traffic
- Off-peak demand

b. Observed Improvements

1. Reduced bus bunching
2. Balanced load distribution across routes
3. Reduced idle bus time
4. Improved passenger satisfaction (waiting time reduction)

The adaptive cost function allowed the system to dynamically respond to traffic spikes and passenger surges.

VII. System Implementation

a. Overall Architecture

The system is implemented using the MERN stack:

Frontend: React.js

Backend: Node.js + Express.js

Database: MongoDB

Optimization Module: HADSA integrated within backend

b. Frontend Implementation (React)

Features:

- Admin Dashboard
- Route Creation Panel
- Bus Allocation Interface
- Real-time Analytics View

Technologies Used:

- React Hooks
- Axios for API calls
- Chart.js for analytics

c. Backend Implementation (Node.js & Express)

Backend modules:

1. Route Manager
2. Optimization Engine
3. Demand Analyzer
4. Scheduling Controller

APIs:

- POST /route/add
- GET /route/optimize
- POST /bus/allocate
- GET /analytics/report

The optimization engine executes:

1. Multi-factor cost calculation
2. Modified Dijkstra path computation
3. Demand ratio evaluation
4. Priority-based bus allocation

d. Database Design (MongoDB)

Collections:

Routes Collection

- routeId
- stops[]
- distance
- trafficIndex

Buses Collection

- busId
- capacity
- currentRoute
- status

Passenger Logs

- stopId
- timestamp
- passengerCount

The NoSQL design ensures scalability and fast read/write operations.

e. Real-Time Adaptability

The system recalculates scheduling at fixed intervals (e.g., every 10 minutes) or when significant demand spikes are detected.

This ensures:

- Near real-time responsiveness
- Reduced congestion impact
- Efficient bus redistribution

VIII. Implementation Advantages

- Scalable architecture
- Low infrastructure dependency
- Real-time adaptive scheduling
- Suitable for smart city deployment
- Cost-effective compared to IoT-heavy solutions

IX. Conclusion

This research presented a Hybrid Adaptive Demand-Based Scheduling Algorithm (HADSA) for intelligent bus scheduling and route optimization in metropolitan public transportation systems. The proposed framework integrates a multi-factor cost function with a modified shortest-path algorithm to dynamically optimize routes based on distance, traffic intensity, passenger demand, and waiting time. Unlike conventional static scheduling approaches, the system adapts to real-time variations in urban mobility conditions.

The algorithm combines route optimization and adaptive bus allocation within a unified decision-making framework, ensuring balanced fleet utilization and improved operational efficiency. Simulation-based performance evaluation demonstrated significant improvements over traditional static and Genetic Algorithm-based scheduling models. The proposed system achieved a substantial reduction in passenger waiting time, improved route efficiency, reduced fuel consumption, and enhanced bus utilization rates.

The MERN stack-based implementation ensures scalability, maintainability, and real-time responsiveness, making the system suitable for deployment in large metropolitan transport networks such as Delhi Transport Corporation.

Overall, the research demonstrates that integrating adaptive scheduling with intelligent optimization techniques can significantly enhance the performance and sustainability of urban bus transportation systems.

X. Future Scope

Although the proposed system demonstrates promising results, several enhancements can be explored in future research:

1. **Real-Time IoT Integration:**
Integration of GPS-based real-time tracking devices to improve traffic and vehicle monitoring accuracy.
2. **Reinforcement Learning-Based Optimization:**
Implementation of reinforcement learning models to automatically adjust weight coefficients (α , β , γ , δ) in the cost function for continuous self-improvement.
3. **Cloud-Based Deployment:**
Migration to microservices architecture using cloud platforms for large-scale deployment and scalability.
4. **Electric Bus Energy Optimization:**
Extension of the model to optimize battery usage and charging schedules for electric bus fleets.
5. **Mobile Passenger Application:**
Development of a passenger-facing mobile application to provide real-time bus arrival predictions and crowd-level information.
6. **Integration with Smart City Infrastructure:**
Collaboration with smart traffic signal systems for coordinated congestion control and route prioritization.
7. **Real-World Dataset Validation:**
Future work may involve validating the proposed model using real operational data from metropolitan bus networks to further evaluate performance robustness.

XI. References

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